

Assignment :- 2 (C4 - Software)

* Scenario :- E-Commerce Platform Using Microservices and Event-Driven Architecture

In a hypothetical e-commerce platform, we adopt microservices to separate core functionalities such as user service, Product service, order service, payment services, and Notification service. Each service operates independently, communicates via REST APIs or messaging queues (eg. Kafka), and is scalable on demand.

Event-Driven Architecture enables these services to communicate asynchronously. For instance :-

- order service emits an "order placed" event.
- Payment service listens and processes payment.
- Notification service listens and sends Confirmation to the user.

* Application of SOLID principles :-

Single responsibility : each microservice handles one specific task

"Open / closed : services can be extended without modifying existing code

Liskov Substitution : interfaces used in services allow easy swapping of implementations

Interface Segregation : services expose minimal, specific interfaces.

Dependency Inversion : event bus or message broker decouples services.

• Observing DRY and KISS : *

• DRY

- Central logging and authentication services reused across microservices
- shared validation libraries.

• KISS:

- use of lightweight message formats (e.g. JSON)
- clear, simple service boundaries and API contracts.